CHS Computation Logic

Community Health Score — Analytics Logic & Formula Reference

Purdue BAIM × Microsoft · Viva Engage Blueprint · Spring 2026

Purpose

This document explains the analytics logic and formulas used to compute the Community Health Score (CHS). It translates technical metrics into plain-English explanations that HR leaders, community managers, and business stakeholders can understand and act on.

Data Sources

GitHub Discussions API: 626 posts/replies across 9 open-source repositories (vercel/next.js, dotnet/runtime, sveltejs/svelte, and more).

StackExchange SEDE: 12,976 questions/answers across 13 technology communities (Python, JavaScript, Docker, Yammer, and more).

1. What Is the Community Health Score?

Five vital signs · One score · Weights adjust by community type

Think of the CHS like a doctor's full health check — not just one test, but five vital signs measured together. A community that has thousands of posts but nobody ever replies is **not** healthy. A community where one person does all the work is fragile, not vibrant. The CHS captures the full picture.

#	Theme	Plain English Meaning	Default Weight	Key Question
T1	Engagement & Activity	Is anyone showing up and participating?	20%	Are people actually here?
T2	Responsiveness & Dialogue	Do posts get answers fast enough to matter?	25%	Does anyone care when you post?
T3	Content Quality & Sentiment	Is the content useful, or just noise?	25%	Is this helping people?
T4	Community Vitality	Is the community growing and renewing itself?	30%	Will this community survive?
T5	Risk & Fragility	Are there warning signs we need to act on?	Penalty	What could break this community?

Master Formula: CHS = (0.20 × T1) + (0.25 × T2) + (0.25 × T3) + (0.30 × T4) + T5_Penalty

Weights shift by community type — a tech community weights speed (T2) more; a support community weights warmth (T3) more.

The Risk Penalty (T5) is always subtracted regardless of community type.

2. How Did We Compute Each Metric?

Every formula mapped to its exact CSV column

We mapped every CHS formula to actual columns in the GitHub and StackExchange datasets. Below is the full translation — what we set out to measure, the formula applied, and exactly which CSV column provided the data.

2.1 Theme 1 — Engagement & Activity

Engagement answers: is anyone actually here? We measure three things — how many unique people participated, whether replies are flowing back to posts, and whether multiple voices join each conversation.

Metric	What We Measured	Formula Applied	Data Source in CSV
Active User Rate (AUR)	% of members who posted or replied at least once	(Unique authors / Total authors in community) × 100	author column — count distinct values
Replies-to-Post Ratio (RPR)	For every original post, how many replies came back?	Total replies / Total posts — normalized over 3.0	is_post=False rows / is_post=True rows
Unique Contributors per Thread (UCT)	Average number of different people joining each thread	Average of unique author counts grouped by post_id	post_id + author — grouped and counted
Gini Coefficient	Is participation spread out or dominated by a few?	Standard Lorenz-curve Gini on per-author contribution counts	author column — contribution count distribution

2.2 Theme 2 — Responsiveness & Dialogue

Responsiveness answers: does anyone care when you post? The fastest way to kill a community is to let questions go unanswered for days. We measure how fast replies arrive and how deep conversations go.

Metric	What We Measured	Formula Applied	Data Source in CSV
Median Time to First Reply (MTFR)	How long does it typically take to get any response?	Median of time_to_first_reply_hrs across posts with ≥1 reply	time_to_first_reply_hrs / time_to_first_answer_hrs column
24-Hour Coverage (24CR)	What % of posts got at least one reply within a day?	Count of replied_within_24h=True / total posts × 100	replied_within_24h boolean column
Thread Continuity Rate (TCR)	What fraction of threads go deeper than one reply?	(Posts with num_replies ≥ 2) / Total posts × 100	num_replies column — threshold filter

2.3 Theme 3 — Content Quality & Sentiment

Content quality asks: is what people share actually useful? A community full of posts that go nowhere, or where nobody marks a question as solved, is failing its members even if it looks active.

Metric	What We Measured	Formula Applied	Data Source in CSV
Helpful Thread Ratio (HTR)	What % of questions resulted in an actual answer?	is_answered=True / total posts × 100	is_answered boolean column
Average Reply Richness (ARR)	Are replies substantive, or just 'thanks'?	Average body_length of replies + 0.3 bonus per has_link/has_code — scaled to 100	body_length, has_link, has_code columns on reply rows

Metric	What We Measured	Formula Applied	Data Source in CSV
Sentiment Balance Score (SBS)	Is the overall tone constructive or toxic?	Set to neutral 60/100 — requires GPT-4 classification in production	Placeholder: would use body text with GPT-4 API

2.4 Theme 4 — Community Vitality & Sustainability

Vitality asks: will this community still exist in 6 months? A community can look healthy today but be quietly dying if no new people are joining, existing members are leaving, and the same topics are recycled week after week.

Metric	What We Measured	Formula Applied	Data Source in CSV
New Contributor Rate (NCR)	Are newcomers making their first post?	(Posts where is_first_contribution=True) / Total posts × 100	is_first_contribution boolean column
Retention Ratio (RR)	Do members come back across multiple time periods?	% of authors who appear in more than one year_week — proxy for return visits	year_week + author — grouped to find repeat appearances
Topic Churn & Novelty (TCNI)	Are fresh topics emerging, or same old conversations?	Set to 50/100 — requires BERTopic clustering in production	Placeholder: requires NLP clustering of body text
Lurker-to-Contributor Rate (LCCR)	Are people who were silent starting to speak up?	Set to 30/100 — requires longitudinal tracking across time windows	Requires cross-period user tracking not in current dataset

2.5 Theme 5 — Risk & Fragility Signals

Theme 5 works differently — it is a penalty system, not a positive score. Think of it as a smoke detector. The louder the alarm, the more points are deducted from the final CHS.

Metric	What We Measured	Formula Applied	Data Source in CSV
Ghost Post Rate (GPR)	% of posts that were completely ignored by everyone	(Posts with is_ghost_post=True after 72 hrs) / Total posts × 100	is_ghost_post boolean column
Ghost Post Penalty	Points deducted based on severity of ignored posts	GPR × 0.5 (max 25 pts) + 10 extra pts if GPR > 40%	Computed from GPR
Super-User Dependency (SUDI)	Is one person doing everything? What if they leave?	% of total content contributed by top 5% of authors	author contribution counts — top 5% share
SUDI Penalty	Points deducted based on fragility level	SUDI 50–65% = –5 pts | 65–80% = –12 pts | >80% = –20 pts	Computed from SUDI

3. What We Can Compute Now vs. What Needs GPT-4 / BERTopic

Honest scoping — designed for production, constrained by budget

The framework was designed to be fully computable. Two components — sentiment analysis (SBS) and topic modeling (TCNI) — require GPT-4 and BERTopic respectively, which were scoped out due to API cost constraints. The architecture fully supports them when budget allows.

Theme	Computable from CSV Now	Needs GPT-4 or BERTopic
T1 Engagement	AUR, RPR, UCT, Gini — all computed from author/post counts	None — all T1 metrics fully computable
T2 Responsiveness	MTFR, 24CR, TCR — computed from timestamps and reply counts	Reply Reciprocity Rate (RRR) — needs user-pair tracking
T3 Content Quality	HTR (is_answered), ARR (body_length + has_link/has_code)	Sentiment Balance Score (SBS) — needs GPT-4 classification
T4 Vitality	NCR (is_first_contribution), RR (repeat authors by week)	TCNI (needs BERTopic), LCCR (needs longitudinal tracking)
T5 Risk	GPR (is_ghost_post), SUDI (top-5% contribution share)	Toxic Sentiment Rate (TSR) — needs GPT-4 classification

Note on CDS (Contributor Depth Score): AUR was replaced by CDS in the dashboard.

AUR was retired because our scraped dataset contains only active users — making AUR trivially 100% for every community.

CDS = Avg contributions per author, normalized so 3.0 avg = score 100. This produces meaningful variation (observed range: 1.22–2.77) and is a stronger loyalty signal.

4. Key Findings Across 22 Communities

What the data actually revealed

4.1 The Ghost Post Crisis

The single biggest issue across all communities is ghost posts — questions or discussions that get zero replies. When someone posts and hears nothing, they stop posting. They stop coming back. They tell others the community is dead.

vercel/next.js (GitHub): 88% ghost post rate — 88 in every 100 posts are completely ignored. CHS = 0/100.

moby/moby (GitHub): 80% ghost post rate. The community has essentially stopped functioning as a two-way conversation.

stackoverflow/docker: 41.7% ghost post rate — just above the 40% threshold where an emergency penalty kicks in. CHS = 5.9/100.

Framework recommendation: Set a team norm that no post goes unanswered for >48 hours. Even 'I'm looking into this' counts.

4.2 Activity ≠ Health — The Super-User Dependency Risk

In moby/moby, the top 5% of contributors are responsible for 37.8% of all content. In vercel/next.js, if the top contributors took a two-week vacation, the community would effectively go dark. This is the 'Autopsy of Friendster' problem — communities that depend on a small number of stars collapse catastrophically when those stars leave.

Communities with SUDI >65% (top 5% generating >65% of content) should immediately identify and train backup contributors.

The community must not be one resignation away from collapse.

4.3 Top Performers — What We Can Learn

dotnet/runtime scores the highest CHS in the dataset. It achieves this with a low ghost post rate (8%), strong engagement diversity (T1=79.4), and solid responsiveness. Its Gini coefficient of 0.393 shows contributions are reasonably spread across participants, not concentrated in a few hands.

sveltejs/svelte is the second-best performer with the best responsiveness score — replies come faster than almost anywhere else. This is achieved by a culture of quick acknowledgments even before full answers are ready.

5. Priority Risk Register

Highest-impact interventions ordered by severity

This table maps directly to the 41-nudge system in the CHS framework. The three communities below the score of 35 are in the critical zone where recovery becomes exponentially harder the longer action is delayed.

Priority	Community	CHS	Key Risk Signal	Recommended Action
CRITICAL	vercel/next.js (GitHub)	0	88% Ghost Post Rate — nearly every post is ignored	Immediate leadership intervention; full community revival program
CRITICAL	moby/moby (GitHub)	5.2	80% Ghost Post Rate + high SUDI (37.8%)	Set team norm: zero ghost posts; train 3 backup experts

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Priority	**Community**	**CHS**	**Key Risk Signal**	**Recommended Action**
CRITICAL	stackoverflow/docker (SE)	5.9	41.7% Ghost Post Rate exceeds 40% emergency threshold	Re-engagement campaign with fresh content; designate on-call experts
WARNING	stackoverflow/python (SE)	22.4	36.8% Ghost Post Rate + 22.3% SUDI	Reply to 3 oldest unanswered posts; acknowledge within 4 hrs
WARNING	stackoverflow/javascript (SE)	29.0	28.2% Ghost Post Rate rising	Commit to daily check-ins; recruit co-moderators
OPTIMIZE	dotnet/runtime (GitHub)	50.2	Performing well — moderate ghost rate 8%	Document practices; raise retention target for next quarter

6. Glossary — Plain English Definitions

For leaders and managers who want the metrics without technical detail

Ghost Post	A post that got zero replies after 72 hours. The poster's voice was completely ignored.
Super-User Dependency (SUDI)	How much of the community's activity comes from just a handful of people. High SUDI = fragile community.
Gini Coefficient	A measure of inequality in who's contributing. 0 = everyone contributes equally. 1 = one person does everything.
Contributor Depth Score (CDS)	Average contributions per author, normalized so 3.0 = score of 100. Replaced AUR in the dashboard.
Replies-to-Post Ratio (RPR)	For every post made, how many replies came back. Below 1.0 means most posts are broadcast into a void.
Median Time to First Reply (MTFR)	How long the typical post waits before getting any response. The lower the better.
24-Hour Coverage (24CR)	What fraction of posts received at least one reply within 24 hours — a proxy for community reliability.
Helpful Thread Ratio (HTR)	What % of question-threads ended with an actual answer or resolution signal.
New Contributor Rate (NCR)	What fraction of posts this period came from first-time posters — a signal of community growth.
Retention Ratio (RR)	What % of last period's contributors came back this period. High retention = healthy, loyal community.
Topic Churn & Novelty (TCNI)	Are new topics emerging each period? Low TCNI = stagnation. Requires BERTopic in production.
Sentiment Balance Score (SBS)	Is the overall tone constructive or toxic? Requires GPT-4 classification in production.

"Community health is not an accident — it is the result of intentional design, measurement, and action."

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